

ShopSphere

S4 - Product Catalog, Variants and Media

Backend + DB

Platform-Specific Sprint Planning Document

This document intentionally defines only the behavior and delivery scope of the selected platform. It does not mix implementation details from the other platforms.

1. Sprint Definition

Sprint: S4 - Product Catalog, Variants and Media

Platform: Backend + DB

Application: ShopSphere

Objective: Implement the backend and database boundary for Product Catalog, Variants and Media, including business rules, persistence and API behavior.

This sprint is complete only when the stated use cases, acceptance criteria and tests pass for this platform.

2. Business Use Cases

ID	Use Case	Actor	Precondition	Main Flow	Expected Result
UC-01	Create product	Admin	Authorized and valid category	Enter product data and variants, save draft/publish	Valid product is persisted and can become active.
UC-02	View product	Customer	Product is active	Open product detail	Customer sees current safe product data.

3. Detailed Implementation Plan

1. Create catalog tables and lifecycle states.
2. Implement product/variant CRUD and validation.
3. Enforce SKU uniqueness, category relationship and publish prerequisites.
4. Expose customer-safe product/detail APIs.

4. Data / State / Contract Scope

- brands
- products
- product_variants
- product_images
- product_attributes

API / Integration Contract

- Feature APIs for this sprint, versioned under /api/v1.
- Every endpoint must define authentication, authorization, validation, error codes and request correlation.

5. Rules and Expected Behavior

- The platform must rely on the backend as the source of truth for business rules.
- Do not expose stack traces, secrets or internal implementation details to end users.
- Handle loading, empty, validation and error states explicitly.
- Requests that can be retried must be safe from duplicate side effects where the feature requires idempotency.
- Ownership and authorization must be checked server-side for protected data.

6. Acceptance Criteria

- Data model supports the required business relationships and constraints.
- API enforces authentication, authorization, validation and ownership where applicable.
- Business rules are server-authoritative.
- Transactions/idempotency/concurrency controls are used where the feature requires them.
- Controlled error codes and request correlation are implemented.

7. Test Cases

TC	Scenario	Expected Result
TC-S4-01	Duplicate SKU	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S4-02	Negative price	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S4-03	Invalid discount	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S4-04	Missing field	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S4-05	Invalid category	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S4-06	Duplicate variant	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S4-07	Inactive product	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S4-08	Unavailable variant	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.
TC-S4-09	Broken image reference	Operation is handled according to the defined business rule; no incorrect state or security leak is produced.

8. Sprint Exit Checklist

- Implementation completed only for this platform.
- All primary and negative use cases pass.
- API contracts are implemented/consumed as defined.
- Validation and authorization tests pass.
- Loading/empty/error handling is verified where applicable.
- No critical or blocker defect remains.
- Automated tests pass in CI.
- Documentation and contract changes are committed.

9. Out of Scope for This Document

- Implementation behavior of Customer Portal. That behavior belongs in the separate S4 document for that platform.
- Implementation behavior of Admin Portal. That behavior belongs in the separate S4 document for that platform.